

Development of an Autonomous Wireless Charging System for Unmanned Aerial Vehicles

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Abstract—Autonomous Unmanned Aerial Vehicles (UAVs) are capable of revolutionizing the delivery industry but suffer from insufficient flight range due to battery limitations and charging requirements. Wireless power transfer offers a solution by creating autonomous recharging stations that can extend the range of UAVs. This paper presents a 700 W fully autonomous wireless charging system with a motorized coil alignment mechanism. Surveying the literature, the level of integration of the system presented is rare; hence, the paper provides a comprehensive design methodology. The system achieves greater than 86% DC-to-DC efficiency for the full constant-current constant-voltage charging profile of two paralleled 7000 mAh UAV LiPo batteries. The alignment mechanism ensures lateral displacement of less than 1 cm, within a search area of 40 cm by 40 cm on the landing platform, allowing maximum power transfer.

Index Terms—WPT, IPT, UAV, autonomous, CC-CV, LCC-S compensation, coil alignment

I. INTRODUCTION

Fully autonomous UAVs have utility in various applications, including package delivery, surveillance, and photography. One of the major limitations of UAVs is the flight duration, constrained by the limited battery capacity and charging requirements [1]. Autonomous docking stations that employ Wireless Power Transfer (WPT) can be an effective method to charge batteries repeatedly without human interference [2]. A stationary pad can facilitate consecutive local missions within range, or even charging one drone while another is in the air for continuous coverage. Deployment of multiple docking stations along a flight path can significantly increase flight duration and enable longer-range deliveries.

Designing Inductive Power Transfer (IPT) based docking stations usually requires a large transmitter coil or an alignment mechanism due to the limited landing precision of UAVs. Some alignment mechanisms for these systems consist of landing correction via computer vision [1], or unique pad geometry [3], Unmanned Ground Vehicles (UGVs) with embedded coils [1], and landing pads with robotic coil alignment [4]. Most IPT chargers proposed in the literature are low power (<150 W) [3], and focus on misalignment tolerance and proof-of-concept. Authors in [5] have developed a 136-W, 80-% efficient WPT charger with large misalignment tolerance. However, the proposed design requires significantly large transmitter coils. In [4], authors proposed a 93% efficient,

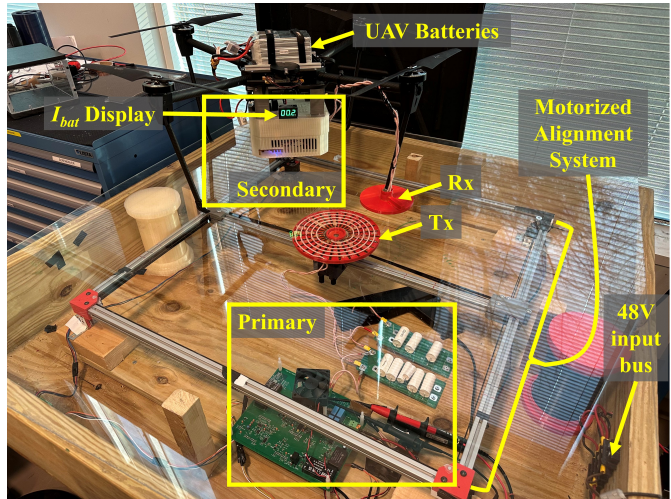


Fig. 1. Complete autonomous drone IPT system.

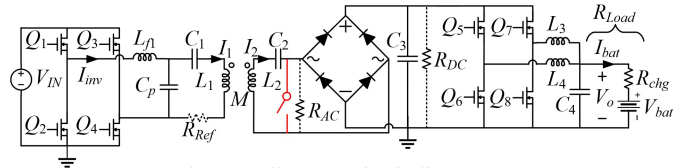


Fig. 2. Full system circuit diagram.

144-W IPT platform with motorized alignment. However, the proposed alignment mechanism has not been experimentally demonstrated. In [3], a 520-W frustum-shaped charging pad was built, and the lack of other high-power chargers was emphasized. The major challenge with the proposed design is that it requires high landing accuracy to enable efficient, fast charging of the drone batteries.

In this paper, the authors have designed and demonstrated a 700-W, autonomous WPT charging pad, as shown in Fig. 1. An LCC-Series (LCC-S) resonant topology is used for charging two paralleled 7000 mAh, 22.2 V LiPo batteries. The lightweight drone-mounted secondary converter is attached beneath the drone frame, while the receiver coil is attached to the foot of the landing gear. As per the authors' best knowledge, this is the first presentation of an IPT UAV charging system with this power range and level of integration. The constructed system exhibits reasonable efficiency (>86%) for the entire constant-current constant-voltage (CC-CV) charging profile of the batteries.

TABLE I
HARDWARE COMPONENT VALUES

Parameter	Variable	Value
Input Voltage	V_{IN}	48 V
Inverter Frequency	f_{inv}	150 kHz
Buck Frequency	f_{buck}	125 kHz
Battery Voltage	V_{bat}	22.2-25.2 V
Charging Current	I_{bat}	10-28 A
Tx Coil	L_1	6.01 μ H
Series Inductor	L_{f1}	2.15 μ H
Rx Coil	L_2	5.56 μ H
Buck Inductor 1	L_3	14.537 μ H
Buck Inductor 2	L_4	14.563 μ H
Mutual Inductance	M	3.25 μ H
Tx Coil Compensation	C_1	300 nF
Parallel Capacitor	C_p	500 nF
Rx Coil Compensation	C_2	200 nF
Filter Capacitor	C_3	43 μ F
Output Capacitor	C_4	40 μ F
Charging Resistance	R_{chg}	30 m Ω

II. LCC-S IPT CHARGER

The circuit diagram of the proposed charger is shown in Fig. 2. The system consists of a GaN-based phase-shift (α) modulated inverter on the primary side. The transmitter Tx and receiver Rx subsystems are inductively coupled via LCC-S compensated resonant network. The secondary side consists of a full-wave diode rectifier, a filter capacitor, and an interleaved buck converter. The LCC-S compensation topology and secondary converter were selected to enable practical control of the secondary to follow the CC-CV charging profile of the battery without continuous communication with the primary, which will be explained in Section IV. CC-CV control isolated on the secondary also enables Tx system interoperability for drones with differing battery loads or charging requirements, as each drone Rx system can then regulate its own charging.

The component values from the hardware setup are given in Table I, which were determined using equations (1)-(7) [6] with the battery specifications at a nominal charging rate of 2C (28 A). Equations (3)-(5) describe the equivalent resistances at different stages of the system highlighted in Fig. 2, starting from the equivalent load resistance R_{Load} , buck converter input resistance R_{DC} , rectifier input resistance R_{AC} , and resistance reflected on the primary R_{ref} . R_{ref} is considered resistive due to tuning L_2 and C_2 at the operating frequency in (2). In these equations, the nominal duty cycle of the interleaved buck (D) was selected to be 0.35 to limit both Rx currents and buck input voltage for switches $Q_5 - Q_8$ from Fig. 2. The IPT coupler inductances were selected following optimization guidelines detailed in [7] for Series-Series (S-S) compensation for minimizing system weight. The compensation was then modified to LCC-S compensation using equations (1) and (2). These equations describe the relationship between the coil compensation (L_1, C_1, L_2, C_2) and the series inductor compensation (L_{f1}, C_p), tuned to maintain reasonable Tx coil current $I_{1,rms} < 20$ A at $\alpha = 150^\circ$ and resonant frequency $\omega_{inv} = 2 * \pi * 150$ kHz. ω_{inv} was not optimized in [7], but higher frequencies will be explored in

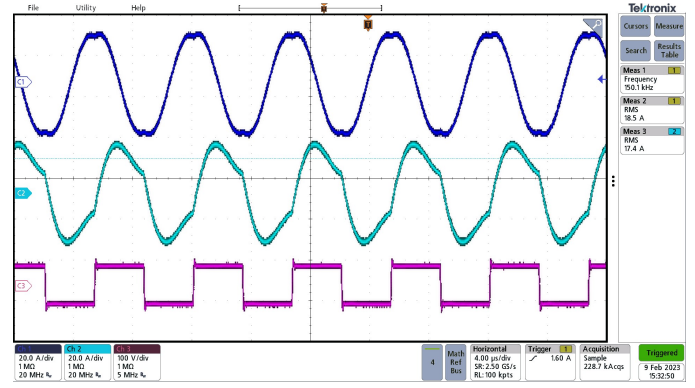


Fig. 3. I_1 (dark blue), I_{inv} (light blue), V_{inv} (pink).

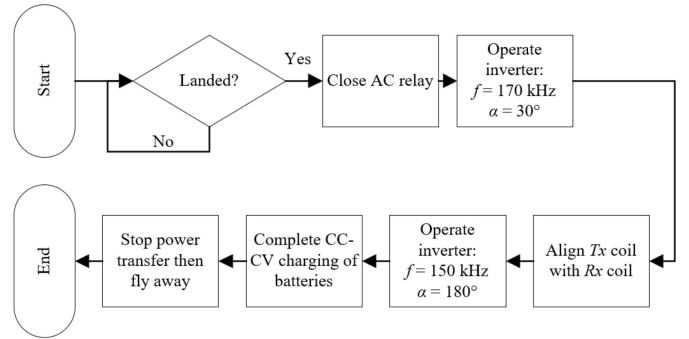


Fig. 4. Charger execution flowchart.

future work. Experimental currents shown in Fig. 3 verify the < 20 A currents from the design, though it should be noted that higher α was necessary to achieve the targeted power due to non-ideal tuning of the resonant network and unconsidered resistances. This system completes autonomous charging by following the execution flowchart shown in Fig. 4.

$$0.20L_1 < L_{f1} < L_1 \quad (1)$$

$$\omega_{inv} = \frac{1}{\sqrt{L_{f1}C_p}} = \frac{1}{\sqrt{(L_1 - L_{f1})C_1}} = \frac{1}{\sqrt{L_2C_2}} \quad (2)$$

$$R_{DC} = \frac{R_{Load}}{D^2} = \frac{V_o}{I_{bat}} \frac{1}{D^2} = \frac{1}{D^2} \left(R_{chg} + \frac{V_{bat}}{I_{bat}} \right) \quad (3)$$

$$R_{AC} = \frac{8}{\pi^2} R_{DC} = \frac{8}{\pi^2} \frac{V_{bat}}{I_{bat}} \frac{1}{D^2} \quad (4)$$

$$R_{ref} = \frac{\omega_{inv}^2 M^2}{R_{AC}} = \frac{\pi^2 \omega_{inv}^2 M^2 D^2 I_{bat}}{8V_o} \quad (5)$$

$$V_{inv,rms} = \frac{4}{\pi\sqrt{2}} V_{IN} \sin\left(\frac{\alpha}{2}\right) \quad (6)$$

$$P = I_{1,rms}^2 R_{ref} = \frac{\pi^2 \omega_{inv}^2 M^2 D^2 I_{1,rms}^2 I_{bat}}{8V_o} \quad (7)$$

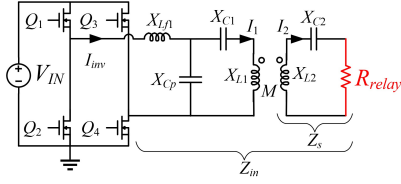


Fig. 5. Equivalent circuit diagram during alignment with shorted rectifier.

III. INDUCTIVE COIL ALIGNMENT

In the proposed alignment system, the coils are aligned by mounting the Tx coil on a motorized 2-axis system, as shown in Fig. 1. A simple, yet highly effective, alignment algorithm based on maximizing I_{inv} measurement is implemented to virtually eliminate misalignment between the Tx and Rx coils within a 40 cm x 40 cm search area. The search area of the motion system was chosen based on the landing accuracy of the drone flight controller and GPS, such that the drone's secondary coil lands within the search area reliably if targeting the center of the pad.

A similar alignment approach is proposed in [4], but for S-S compensation and without hardware implementation. Additionally, in [4], an impedance measurement algorithm is proposed at resonance to be used in alignment, but in practical implementation of S-S, a Tx coil without the presence of Rx coil at resonance will exhibit a short-circuit condition on the primary that must be addressed.

$$\omega_n = \frac{X_L}{jnL} = \frac{1}{jnCX_C} \quad (8)$$

$$Z_{s,n} = jn\omega_a L_2 + \frac{1}{jn\omega_a C_2} + R_{relay} \quad (9)$$

$$Z_{in,n} = jn\omega_a L_{f1} + \frac{1}{jn\omega_a C_p} \quad (10)$$

$$\parallel \left(jn\omega_a L_1 + \frac{1}{jn\omega_a C_1} + \frac{M^2 n\omega_a^2}{Z_{s,n}} \right)$$

$$V_{inv,rms,n} = \frac{4}{\pi n\sqrt{2}} V_{IN} \sin\left(\frac{n\alpha}{2}\right) \quad (11)$$

$$I_{inv,rms} = \sqrt{\sum_{n=1,3,5}^{\infty} \left(\frac{V_{inv,rms,n}}{|Z_{in,n}|} \right)^2} \quad (12)$$

The alignment implemented in this paper is initiated via Bluetooth module mounted on the drone, which first closes the AC relay from Fig. 2 highlighted in red. This shorts the input of the diode rectifier on the secondary system, creating the equivalent circuit shown in Fig. 5. Equations (8)-(10) describe the impedances throughout this equivalent circuit, evaluated at the alignment frequency ω_a . If ω_a is selected to be the resonant frequency, ω_{inv} , (9) reduces to just the small value of R_{relay} , reflecting large impedance to the primary side, assuming maximum M in (10), and creating virtually an open circuit at the Tx branch. This would cause a short circuit condition through the inverter, as X_{Lf1} and X_{Cp} have been tuned at the resonant frequency. To avoid this issue, ω_a was

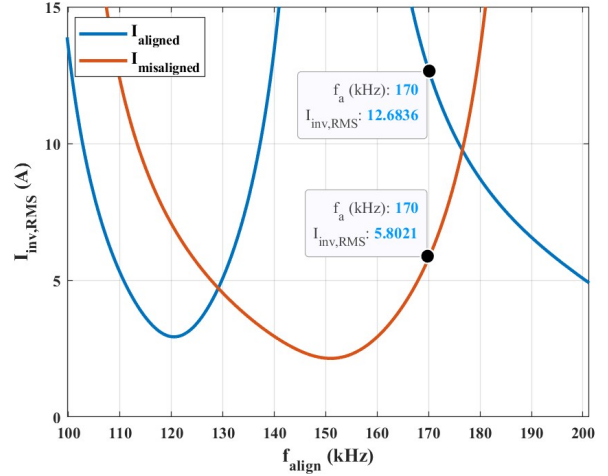


Fig. 6. Analytical inverter current for different alignment frequencies during both perfect alignment and absence of secondary coil.

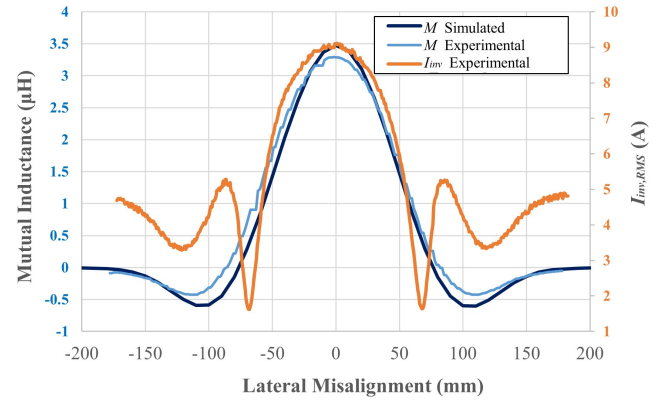


Fig. 7. Current and mutual inductance vs. lateral displacement.

selected based on equation (12), derived from (9)-(11), such that given a constant α in (11), I_{inv} increases as M increases, corresponding to lateral alignment. This relationship of the current allows the alignment algorithm to maximize current measurements to maximize the coupling of the Tx and Rx coils. Selecting $\alpha = 30^\circ$ and substituting the component values from the AC Relay datasheet and Table I into (12) while sweeping values of ω_a yields $I_{inv,rms}$ when aligned, and then gives misaligned current when $M = 0$, corresponding to no mutual coupling between the two coils. Using this approach, analytical $I_{inv,rms}$ is plotted around the resonant frequency in Fig. 6, confirming the short circuit condition at resonance when the Tx and Rx coils are aligned. The selection of the alignment frequency is important because it determines the gradient of the current that the alignment algorithm uses to maximize current. From this analysis, 170 kHz was selected for the alignment frequency to maximize I_{inv} with M , to maintain an inductive inverter load, and to limit the maximum current possible during alignment. At this frequency, the experimentally measured relationship between I_{inv} and M vs. lateral displacement is shown in Fig. 7. This

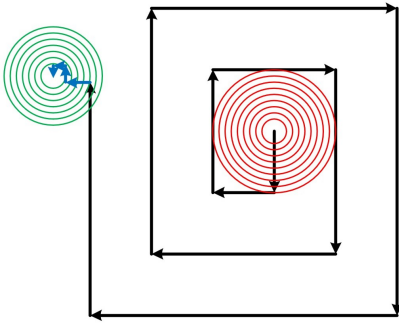


Fig. 8. Typical transmitter coil (red) movement to find the secondary coil (green).

characterization was captured by moving the Tx coil in small increments across the Rx coil, while measuring the M via induced open circuit voltage. The measured peak current is notably smaller than in Fig. 6 due to resistances and parasitics unaccounted for in the impedance analysis above.

Following the flow chart in Fig. 4, after closing the relay, the Bluetooth module signals the primary side to start the inverter for alignment. While the inverter is operating at 170 kHz and 30° phase-shift, the primary coil spirals outward from the center of the landing platform while sampling I_{inv} . Once current is measured above a calibrated threshold, indicating the secondary coil is within sensing range, the algorithm maximizes current along x-y axes twice for reliability. Typical Tx coil motion for this process is depicted in Fig. 8, where the black arrows are the spiral and the blue arrows are x-y current maximization.

This process can take up to 1 minute if the Rx coil lands far from the center. The detection time could be decreased significantly with mechanical improvements to the 2-axis motion system, as the main limitation was the speed at which the coil could be moved smoothly while sampling current. At higher speeds, the coil would shake while spiraling, which could affect current measurements and unnecessarily stress the motors. Alternatively, different coil geometry or size could increase misalignment tolerance as seen in the literature [5], giving a larger area to detect the Rx coil sooner, but this may inhibit the maximum power or efficiency of the system. As the coil alignment process occupied <5% of the intended charging duration, further optimization of the detection speed was not pursued.

After the system finishes the alignment process, the inverter turns off and switches back to the resonant frequency ω_{inv} to prepare for charging.

IV. CONTROL OF IPT SYSTEM

The control approach envisioned for this system is based on enabling full control of CC-CV charging on the secondary system without requiring slow and fallible wireless communication with the primary controller. To achieve this, the Tx coil current (I_1) is held at constant magnitude, such that the induced voltage magnitude at the secondary (V_M) is constant, following equation (13) from [8]. This induced

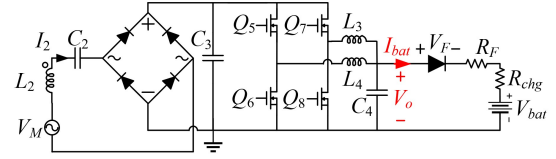


Fig. 9. Equivalent model for controller plant derivation.

voltage supplies the circuit in Fig. 9 used to derive the state equations given in (14), which will be discussed later in this section. LCC-S compensation topology is selected for this charger over other resonant topologies because it can maintain a stiff primary coil current throughout all operating points. LCC has this current source ability because L_1 and C_1 are detuned at ω_{inv} from (2), adding reactance to the Tx coil branch and buffering load changes reflected from the secondary, evident in equation (10). Thus, it can maintain stiff current operation without a controller given that the magnitude of the voltage across the resonant tank, $V_{inv,rms}$, is held constant.

$$V_M = j\omega M I_1 \quad (13)$$

$$\begin{cases} \frac{dI_2}{dt} = \frac{1}{L_2} (V_M - V_{C2} - V_{AC} - I_2 R_{L2}) \\ \frac{dV_{C2}}{dt} = \frac{I_2}{C_2} \\ \frac{dV_{C3}}{dt} = \frac{1}{C_3} (I_{DC} - D I_{L3} - D I_{L4}) \\ \frac{dI_{L3}}{dt} = \frac{1}{L_3} (D V_{C3} - V_{C4} - I_{L3} R_{L3} - I_{L4} R_{L4}) \\ \frac{dI_{L4}}{dt} = \frac{1}{L_4} (D V_{C3} - V_{C4} - I_{L3} R_{L3} - I_{L4} R_{L4}) \\ \frac{dV_{C4}}{dt} = \frac{1}{C_4} (I_{L3} + I_{L4} - \frac{V_{C4} - V_F - (I_{L3} + I_{L4}) R_F - V_{bat}}{R_{chg}}) \end{cases} \quad (14)$$

$$V_{AC} = \frac{4}{\pi} V_{C3} \frac{I_2}{\|I_2\|} \quad (15)$$

$$I_{DC} = \frac{2}{\pi} \|I_2\| \quad (16)$$

Other topologies could be implemented using a controller to maintain the primary current constant. For example, S-S compensation was tested in this design before LCC-S because of its higher efficiency under full load. The S-S coil current was maintained constant using a high-bandwidth controller to decouple its dynamics from the Rx control, but the low quality factor of the lightweight secondary system resulted in harmonic Tx coil current that interfered with controller stability. The LTSpice current of the S-S implementation at 700 W output is shown in Fig. 10 for reference, with spectrum analysis results shown in Fig. 11, highlighting the contribution from harmonics. The higher harmonics carried more current as power increased beyond 500 W, decreasing the peak of

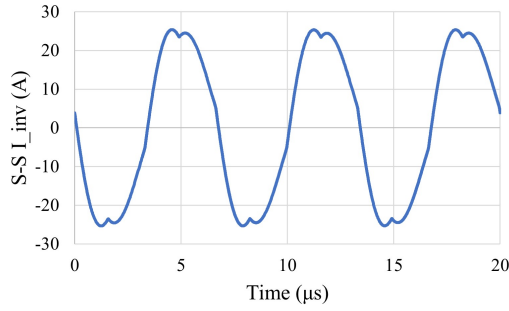


Fig. 10. Simulated S-S transmitter coil current.

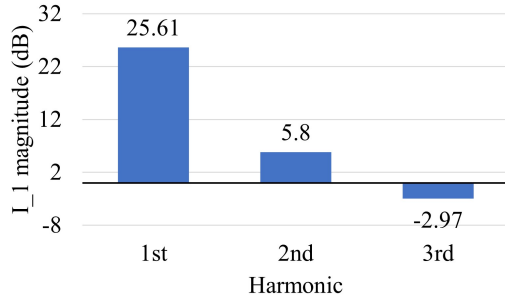


Fig. 11. Simulated S-S transmitter coil FFT results.

the current while increasing the RMS value. Thus, the peak-detector circuit that was used for current sensing produced non-linear measurements, shown in Fig. 12, which destabilized the primary current control. This issue is circumvented by modifying the compensation to LCC-S leveraging the benefits mentioned in the previous paragraph. LCC-S does not achieve as high efficiency as S-S under loaded conditions, but the modification was practical to achieve controller stability without redesigning the PCB.

On the secondary, the interleaved buck duty cycle is modulated with dual-loop control. The outer loop controls V_o and the inner loop controls I_{bat} , following a CC-CV charging curve at a CC charging rate of 28 A up to the CV voltage of 25.2 V. Proportional-Integral (PI) controllers are frequently used for this converter and were selected for both loops for ease of implementation. Extended Describing Function (EDF) is a common method for modeling resonant DC-DC converters

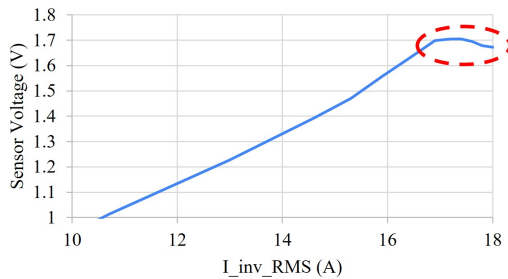


Fig. 12. Experimental sensor calibration readings.

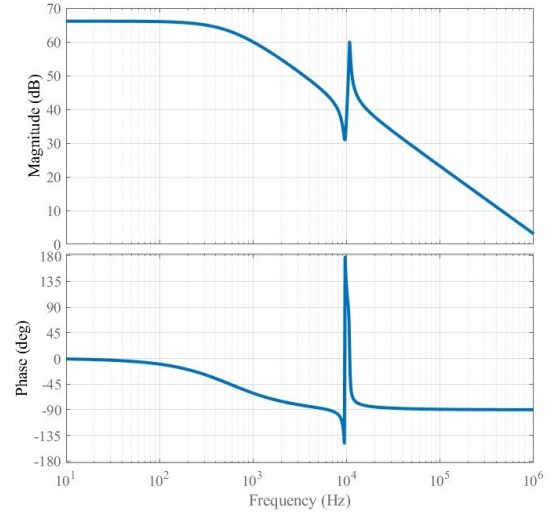


Fig. 13. Bode plot of EDF-derived transfer function from duty cycle D to I_{bat} for the inner current loop.

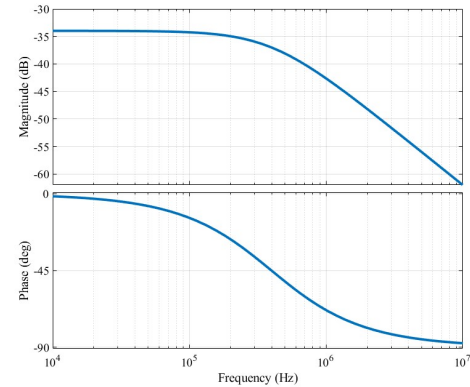


Fig. 14. Bode plot of EDF-derived transfer function from I_{bat} to V_o for the outer voltage loop.

[9] [10], and was initially used to model the AC-coupled interleaved buck dynamics as it is accurate for both AC and DC circuits. The state equations in (14) were derived using KVL and KCL on the equivalent model in Fig. 9. Equations (15) and (16) embody the full-wave rectifier in continuous-conduction mode (CCM). The rectifier output current I_{DC} is the average of the rectified sinusoidal input current and the voltage reflected to the rectifier input V_{AC} is assumed to be a square wave in phase with the input current. I_2 and V_{C2} are the AC state variables, which get decomposed into d and q components through EDF, yielding 8 total state variables. The state matrix is formed from this decomposition, while the input variable is just D . Linearizing the system using Jacobian matrices yields a state-space to generate the plant transfer functions shown in the bode plots in Fig. 13 and Fig. 14. The resulting PI controller crossover frequencies were designed around 30 Hz for the outer V_o loop and 300 Hz for the inner I_{bat} loop to ensure decoupling between the two.

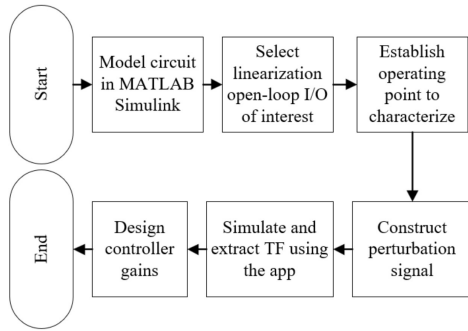


Fig. 15. Simplified controller design flowchart.

Unfortunately, the lightweight design of the secondary system forced small inductance of the IPT coupler, and consequently low quality factor. The poor harmonic attenuation made it difficult to design controllers using EDF, as it is based on the First Harmonic Approximation (FHA). At low power operating points <300 W, the rectifier also enters discontinuous conduction mode (DCM), strengthening the harmonics, slowing the current controller bandwidth, and introducing steady-state error when designed with EDF. As this controller performance occurred in MATLAB Simulink simulation as well as hardware, the issue was addressed by performing frequency response estimation using MATLAB's Model Linearizer application. The model was run at several different low-power operating points measuring I_{bat} , then perturbed at varying frequencies to generate points to estimate the transfer functions. This process is documented by MathWorks and summarized in Fig. 15. For each of these transfer functions, PI gains were designed at a crossover frequency of 300 Hz, generating a lookup table of PI gains corresponding to operating points defined by I_{bat} . These scheduled PI gains were implemented in the Rx microcontroller and interpolated linearly, adjusting the values experimentally to ensure decoupling of the two control loops.

V. EXPERIMENTAL RESULTS

The alignment accuracy was evaluated by placing the Rx coil at various positions marked in green in Fig. 16, then repeating the alignment algorithm several times and recording the M measured at the resulting Tx locations, marked in yellow. These positions were chosen to capture accuracy at different Rx distances from the center of the platform. At each position, a reference M was measured by manually moving the coil until the highest current was seen on the oscilloscope. The difference in M from the reference measurement after running the alignment algorithm can be mapped to lateral displacement from the prior characterization of the IPT coupler in Fig. 7 if normalized. The alignment algorithm could not be expected to reach the reference M every time, as the algorithm has fixed step distances. However, displacement was observed to always be <1 cm, averaging 6 mm; corresponding to $<1\%$ average reduction from the reference M at each coil position. This accuracy is evident in Fig. 16, given how close all yellow markings are to the green target positions.

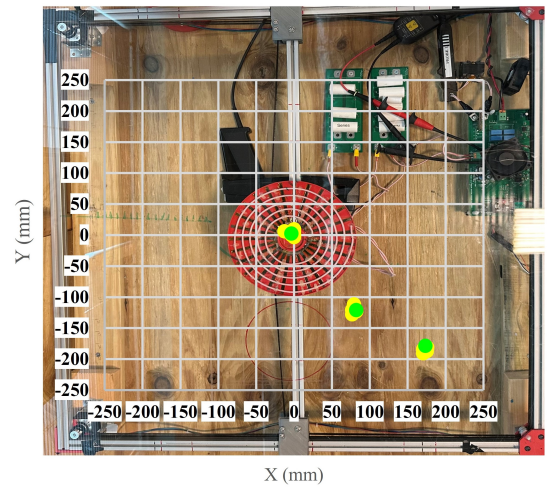


Fig. 16. Recorded alignment attempts. Target (green) Algorithm Result (Yellow).



Fig. 17. Flight test of autonomous charging system.

After finishing the IPT UAV charger, the secondary subsystem was mounted on the underside of one of the UAVs from AERPAW [11], shown again in Fig. 17, for testing the charging procedure. Initially, it was designed for the coil to be directly below the enclosure, but magnetic field interference required changing its position to be farther from the electronics, hence its position on the leg of the drone. The UAV charging system was tested in the field by hovering the drone until the battery had drained enough, then landing the drone and triggering the automated charging process. The alignment system centered the Tx and Rx coils, then the charger took about 15 minutes to recharge the battery, and the drone could fly away. The power level of the field demonstration was halved due to concerns about flying the drone loaded with two batteries and the charger, which ended up weighing 1.4 kg, so the charger operated at 2C for one 7000 mAh battery (14 A).

As designed, LCC-S maintained roughly constant primary current, varying by approximately 1 A over the entire opera-

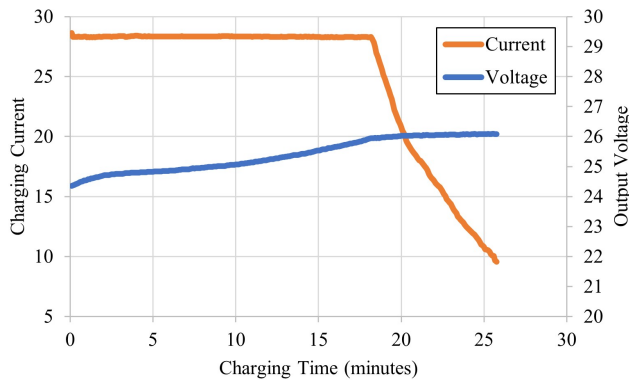


Fig. 18. Experimental CC-CV charging curve.

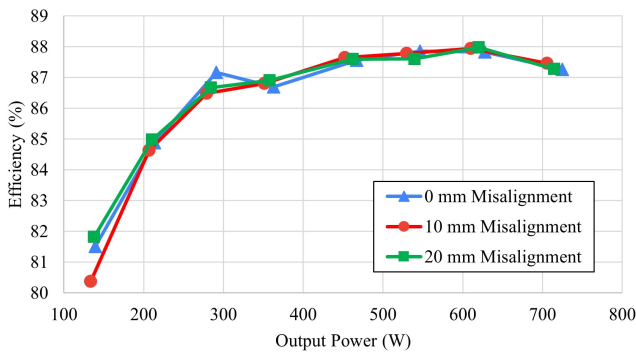


Fig. 19. Experimental efficiency for different lateral displacement.

tion range. The CC-CV charging curve of the LiPo batteries is successfully controlled with the dual loop, shown in Fig. 18. The voltage is shifted up from the specification in Table. I due to the protective diode in front of the battery, as mentioned in the previous section and shown in Fig. 9, having a voltage drop of 0.8 V. The efficiency vs. power is shown for different lateral displacements in Fig. 19, demonstrating the utility of the alignment system that has guaranteed <10 mm misalignment.

VI. CONCLUSION

In this paper, a fully autonomous wireless charging platform was introduced for UAVs. The alignment algorithm eliminates misalignment by ensuring lateral displacement <1 cm. LCC-S resonant compensation was selected for this system to provide robust operation, and the resulting efficiency of the system maintains $>86\%$ for the entire CC-CV charging curve. This paper highlights the challenges in designing lightweight IPT charging systems for UAV applications regarding quality factor and managing harmonic current as well as encountering controller modeling difficulty with DCM. The approaches used in this paper were successful, but careful system design and more tolerant sensing can mitigate the instability of the controllers.

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